Does a meta-doctrine system change model behaviour, or just make it legible? A self-ablation study under equal compute

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Abstract

"Agentic" coding setups increasingly ship a *meta-doctrine layer*: a body of always-on guidance—roles, gates, principles—plus a record that accumulates across sessions, on the premise that this scaffolding improves the work. We ablate one such system (the "Soul System") against the confound that dominates this class of claim: loading doctrine also loads *tokens*, and additional context alone can make a model more thorough. Every apparent gain is therefore re-run against a **length-matched, semantically-empty filler** of equal token budget; a gain that vanishes under this control was compute, not doctrine. Across ten doctrine probes and six instrument probes (Haiku 4.5, Sonnet 4.6, Opus 4.8; $n = 5$–7 per cell), the equal-compute control catches four false positives, and the surviving substance collapses onto a single axis: the system changes the model's *decisions* only when it carries **project-specific knowledge or counter-default conventions the model cannot regenerate and a generic equivalent gets wrong** (record-recall 10/10 vs. 2/5 for a length-matched irrelevant record; convention-transmission flips a counter-default decision 10/10 where generic principles give the default answer 0/10). Everything aimed at generic reasoning-lift—forced roles (as prose *and* as dedicated sub-agents), trajectory-reading, compliance-resistance, the handoff and verification *formats*—is reproduced by the filler. A synthetic multi-session harness extends this to the system's strongest claim: a counter-default fact recorded early and buried under later work prevents a dangerous drift 5/5 $\rightarrow$ 0/5 in a later session, and—uniquely among our results—**does not dissolve at the frontier**: the strong model, lacking the recorded fact, fails *more* confidently, fabricating a non-existent API affordance to justify the unsafe path. The organizing principle is **derivable-dissolves, unguessable-persists**. We give the resulting keep/cut decision and a precise ledger of what a single-shot method cannot settle.

1 Introduction

A growing class of tooling wraps a language model in a *meta-doctrine layer*: standing instructions loaded into every session (personas or "roles," verification gates, design principles) together with a durable project record (decision logs, findings, conventions) that grows over time. The implicit and often-stated promise is that this layer *improves the work*—the model reasons more carefully, catches more defects, and stays coherent across a long project.

This promise is unusually hard to falsify by inspection, because the layer reliably changes the *surface* of the output: longer, more structured, more self-aware-sounding text. That surface change is consistent both with the layer genuinely improving the *substance* of decisions and with it doing nothing but making the model verbose and the work legible. Distinguishing the two requires a control the field rarely applies.

We study one mature instance—an open meta-doctrine system we call the Soul System, comprising ~16 commands, a three-layer always-on doctrine, and an append-only record—and ask a deliberately deflationary question: **which parts change what the model decides, and which only change how the work reads?** Our contributions are:

1. **An equal-compute ablation protocol** (Section 3) that re-runs every apparent win against a length-matched, semantically-empty filler, isolating doctrine-substance from the compute that any long prompt supplies.

2. **A behavioural map** (Section 4) across doctrine and instruments, capability tiers, and task structures, including four wins that the control falsifies.

3. **A single organizing axis**—*derivable-dissolves, unguessable-persists*—that predicts what survives, and a synthetic multi-session result showing the surviving value is the first in the study that does not dissolve at the frontier.

4. **A keep/cut decision** for a lean version (Section 5) and an explicit ledger of what a single-shot method cannot measure (Section 6).

The result is not a defence of the system and not a debunking. It is a separation: a narrow, real, capability-robust value (carrying the unguessable) from a larger body of legibility that a generic equivalent matches.

2 Related work

Context quantity as a confound, and what filler does. The premise of our control is that added tokens can change a model's output independent of their content. The direction of that effect is not simple, and the literature is split in a way that matters for interpretation (Section 6). Pfau et al. [14] show that even *semantically-empty* filler tokens can supply usable parallel computation—small transformers solve problems with filler they cannot solve answering immediately—though the ability is hard to acquire and was absent in commercial models of that era. In the opposite direction, and more robustly, *irrelevant* context **degrades** reasoning: Shi et al. [15] ("distracted by irrelevant context") find accuracy drops sharply when in-topic but irrelevant information is added. So extra tokens are not free, and their sign depends on model and content—which makes a length-matched filler the right instrument but its bias direction something to argue, not assume.

Roles and personas. Assigning static personas in system prompts does not reliably improve accuracy; Zheng et al. [1] test 162 roles across model families and find no general benefit, and Kim et al. [7] show role-play *degrades* reasoning on a third of datasets even at the frontier, with bidirectional, near-zero net effect. The boundary: Kong et al. [8] find an engineered multi-turn *role-immersion* prompt helps zero-shot reasoning—but as an implicit chain-of-thought trigger, a different intervention than a static persona. This matches and bounds our doctrine result (Sections 4.1, 4.3).

Single- vs. multi-agent decomposition. Under matched compute the case for multi-agent orchestration weakens. Tran and Kiela [2] show single-agent LLMs *outperform* multi-agent systems on multi-hop reasoning at equal thinking-token budgets and argue reported multi-agent advantages are "better explained by unaccounted computation"; Gao et al. [3] find the advantage diminishes as base models improve. Notably the canonical multi-agent-debate result [9] reports gains *without* a compute-matched baseline—the exact confound our method targets. Our agentic-roles probe (Section 4.3) reaches the same conclusion for Soul's roles rebuilt as sub-agents.

Knowledge vs. reasoning; retrieval over parametric memory. Our organizing axis—*derivable dissolves, unguessable persists*—restates a robust result. Lewis et al. [10] (RAG) show parametric models have "limited ability to access and precisely manipulate knowledge" and that adding a non-parametric retrieved store beats the parametric model; Ovadia et al. [11] find retrieval "consistently outperforms" unsupervised fine-tuning for injecting facts the model lacks. Gan and Sun [4] (RAG-MCP) extend the same retrieve-don't-preload logic to tool selection, underwriting both our never-load-everything doctrine and the skill-routing probe (Section 4.4).

Calibration and confident fabrication. Our longitudinal result turns on a model *fabricating* a missing fact rather than abstaining. The GPT-4 technical report [12] documents that the model "can be confidently wrong... not taking care to double-check when it is likely to make a mistake," and that RLHF post-training *degrades* calibration (expected calibration error rising roughly tenfold from the pre-trained model). Kadavath et al. [13] show models have partial but imperfect self-knowledge of what they know. We return in Section 6 to what this does and does not license about *scale*.

Positional use of long context. Liu et al. [5] ("lost in the middle") show models use information at the beginning and end of long inputs far better than the middle—the degradation a record-burial probe must control for, and a caveat on our depth result (Section 6).

Self-critique limits. Verification gates assume a model can catch its own errors; Huang et al. [16] find LLMs "cannot self-correct reasoning yet" without external feedback—consistent with our result that the gate *concept* helps but its elaborateness does not (Section 4.2).

Persistent memory and rules. External-memory agents (MemGPT [17]; generative agents with a memory stream [18]) make the cross-session-record hypothesis concrete, though neither isolates record-content from compute as we do. Constitutional AI [6] steers behaviour from a small explicit rule-set rather than dense supervision; the Soul System's compressed always-on rules are a project-level analogue (Section 4.2).

We position this as an *engineering self-ablation*, not a survey: its contribution is the equal-compute discipline applied uniformly and the longitudinal harness, not breadth of coverage.

3 Method

3.1 Arms and models

The unit of measurement is an **arm**: a tuple `(task, condition, model, run-index)`. Each arm is an isolated, non-interactive model call (`claude -p "<task>" < /dev/null`, fresh process, no chat history); the thing under test (doctrine, a role lens, a gate checklist, a handoff cursor, a record) is supplied via the system prompt or pasted into the task, and the bare arm receives the task alone. We run $n \geq 5$ per cell unless noted, across a capability range—Haiku 4.5 (weak), Sonnet 4.6 (mid), Opus 4.8 (frontier)—because the effect size of any aid depends on whether the bare baseline already succeeds.

3.2 The three controls

C1 — Equal-compute control (decisive). The dominant confound in this class of claim is that loading the system also loads tokens. Every apparent win is re-run against a **length-matched, semantically-empty filler** of the same budget: ∼26 KB of unrelated warehouse-logistics prose in place of ∼26 KB of doctrine; a generic "double-check your work" in place of a named gate checklist; a plain summary in place of a structured cursor. *A substance verdict that vanishes under equal compute is not substance.* C1 caught four false positives here (Section 4.1–4.2).

A direction caveat (developed in Section 6). The filler is not a neutral inflator: the literature shows added irrelevant tokens more often *degrade* than improve a model [15], while empty filler can occasionally *supply* compute [14]—both effects are model- and content-dependent. This makes

C1 a **conservative** control rather than a symmetric one: a gain that *survives* the filler is real a fortiori (the filler, if anything, biased against the bare-plus-filler arm), but a gain that *vanishes* is ambiguous between "was compute" and "the filler actively hurt the control arm." Our four filler types are *coherent-but-irrelevant* (warehouse prose) or *generic-nudge* ("double-check"); where they tied doctrine they generally also beat the bare baseline (Section 4.2), so they were not degrading in those cells. We run the clean separation—a three-arm control with a coherent-irrelevant distractor alongside empty padding—in Section 4.6, and it confirms on-model that coherent-irrelevant filler is not neutral while empty padding $\approx$ bare.

C2 — Difficulty validation. A task measures an aid only if the bare baseline plausibly fails it. Before scoring, the bare arm is run $n \geq 5$; if it already succeeds, the task measures ceiling, not the aid, and is discarded or sharpened. A corollary learned the hard way: a high-salience domain (e.g. patient safety) can trigger a caution reflex that *substitutes* for the discipline under test, passing the baseline for the wrong reason; one medical task was discarded for exactly this. Domains are kept low-salience.

C3 — Trap taxonomy. When a task plants a defect, the defect is of a named kind so a "catch" maps to a specific discipline: a false premise (anchoring), a dropped requirement (completeness), a buried downstream consequence (trajectory), a silently-truncated long-horizon constraint (gate). One trap per task.

3.3 Scoring

Rubrics and answer keys are **pre-registered** before outputs are seen. Outcomes are scored mechanically where possible (count the behaviour; within-cell comparable keyword patterns) and then **read-confirmed** on borderline arms. Counts are never eyeballed, and we explicitly watch for *wrong-reason* passes (an artifact refused for missing information, a placeholder flagged, a disaster-salience reflex)—these are designed out or scored separately. One pass in this study *looked* like a decisive win on a keyword count (5.8 vs. 2.6) and reversed on the binary read (both 5/5): the count had measured verbosity of flagging, not catches.

3.4 Three honest value buckets

Not everything is single-shot A/B-testable, and faking a single-shot proxy for a longitudinal or human-workflow claim would be the precise self-consistent-but-false result the study exists to catch. Each component is assigned a bucket: **Measured** (A/B + C1), **Reasoned** (friction/convenience, assessed by transparent cost-vs-value argument, not a faked experiment), or **Inspected** (one-time/structural artifacts).

3.5 Harness and a contamination finding

Arms run from a scratch working directory; condition files are supplied via the model CLI's system-prompt-file flag or pasted inline. **Cross-file `@`-imports are not used** under the non-interactive mode: they fail silently and the model confabulates the missing content $\sim$43% of the time (this is itself one of the project's findings, and the deciding knowledge in the recall probe), so content is inlined and a sentinel load-test confirms it reached the model before any run is trusted. A separate contamination was caught and fixed: non-interactive arms have tool access and read the working directory, so arms launched from a directory containing the answer file simply *grepped it* even in the no-record condition; every arm is therefore run from an **isolated empty directory**, and the contaminated run is retained as evidence.

The harness, the per-experiment pre-registrations and results, and the scoring spec are packaged as a reusable benchmark—the *meta-doctrine ablation suite*—in its own repo `Soul-Benchmark` (extracted from this repo at the 1.0 release), so the experiments in Sections 4 and 6.1 can be re-run and others can plug in their own doctrine and models.

3.6 Known bounds of the method

The harness is single-shot / single-resume; the system's strongest claim is intrinsically longitudinal, reached here by a synthetic session-chain harness (Section 4.5) rather than real project history. n is small (5–7/cell); keyword scoring is within-cell comparable, not absolute. The scorer is also the experimenter; rubrics are pre-registered to constrain this, and where bias is plausible it runs *toward* the system, so null results are conservative.

4 Results

Results at a glance. Every row is run against the equal-compute filler control (C1); a result that vanishes under C1 was tokens, not doctrine. "Survives" = the gain remains under the length-matched control. (Full designs and data: Sections 4.1–4.6; consolidated tables in `data/`.)

Probe	Control	Key result	Survives?	Verdict
Doctrine: roles / gates (frontier)	C1	changes *form* (named, legible), not substance	— (form)	lean to a line
Doctrine: hard regime, weak baseline	C1	substance only on **anchoring** (filler 0/3)	**yes** (narrow)	KEEP anchoring
Record recall (unguessable fact, F038)	C1 + 3-arm	**record 10/10** vs filler $\approx$ 0; isolates cleanly	**yes**	**KEEP — clearest win**
Convention transmission (docs-near-code)	3-arm	flips a counter-default decision; but derivable + primable	**yes** (narrower)	KEEP content, moderated
Roles as sub-agents	C1	A2 $\approx$ C1; multi-agent $\leq$ single; synthesis drops findings	no	form, prose→agents
Skill routing	oracle vs self-pick	**oracle 5/5 vs budgeted 1/5** (weak)	**yes** (bounded)	KEEP tested catalog
Longitudinal carry (7 rungs)	C1, synthetic chain	drift **5/5→0/5**; **persists at the frontier** (fabricates when missing)	**yes**	**KEEP — strongest**
Verify gate (high + low stakes)	3-arm $\times$2	bare ceilings; doctrine adds only a citation (apparent stabiliser did not replicate — n=10 rerun)	no	form / legibility
Handoff cursor	3-arm (cursor vs prose)	**cursor 10/10 = prose 10/10**	no	form / legibility
Calibration (Haiku→Sonnet→Opus)	3-tier ladder	fail-to-recall flat (robust); confident-fabrication *gradient* did **not** reproduce (scoring-fragile)	n/a	non-elimination only (Fig. 1)
Depth + position	token-scale burial	**30/30**, no middle decay (unique needle)	n/a	bounds the depth claim

Derivable dissolves; unguessable persists. The detailed sections follow.

4.1 Doctrine (the always-on layer)

Against a careful frontier baseline, single-role and single-gate ablations changed the *form* of the work (named, legible, auditable) far more than its substance or reliability. An engineered hard regime—a buried logic trap, a false premise under authority pressure, and a ten-constraint long-horizon task—was also form-not-substance at the frontier (the completion-gate ablation on the long-horizon task was null, gate $\approx$ bare).

At a **weak baseline** (Haiku), with the integrated doctrine and C1 throughout, doctrine-attributable substance survives in exactly one place: **challenging an unverified premise** ("anchoring"). It lifted the false-premise task $0/3 \to \sim 1.5/3$ while the filler stayed $0/3$, and replicated on a second anchor task (whole 3/3 vs. filler 0/3). This substance is *content-dependent* (generic thoroughness cannot fake it), *weak-baseline-only* (a mid-capability model self-catches), and *task-structure-gated* (analysis-inviting tasks self-catch earlier than compliance-inviting ones).

Everything else dissolved under C1, including three apparent generalizations: trajectory-reading (bare $1/3 \to$ whole 3/3, but filler 2/3—generic long-context), compliance-flagging (filler reproduced it—disposition), and stakes-gated withholding (filler $5/5 =$ whole—disposition, not doctrine). Without the equal-compute control the study would have claimed a "frontier-ward compliance edge" that does not exist. The consonance with Zheng et al. [1] (roles don't lift accuracy) is direct.

4.2 Instruments and the record

Handoff/resume (Measured; form). A fresh session given a curated record snapshot plus one of {structured cursor, length-matched generic prose summary, terse compact, nothing} surfaced more constraints with any rich note than cold ($\approx +2$ constraints; 0 vs. 1 scope-invention error)—but the *Soul format is form*: a length-matched generic summary tied or beat the cursor (Sonnet 4.8 generic $\geq$ 4.6 cursor; Haiku equal). Richness matters, structure does not.

Record-recall (Measured; KEEP — the clearest win). A trap task whose deciding knowledge is a real, non-obvious project finding was run with {relevant record, no record, length-matched *irrelevant* record (C1)}. The relevant record produced trap-avoidance **10/10**; the irrelevant record **2/5**; no record 2–3/5 (capable arms partially reason toward the right hedge but mostly still recommend the trap). The *specific recorded content*—not "having a record," not the tokens—makes the model reliably correct.

Verification gate (Measured; concept yes, format no). Finalizing a draft with one planted unanchored-claim defect: bare caught it 2/5; *any* check (generic-vague, generic-checklist, or the Soul checklist) lifted it to $\sim$5/5. The gate *concept* earns its place; the Soul-specific checklist does **not** beat a generic "double-check carefully" on catch-rate (both 5/5)—it adds itemized legibility, which is form. (A second, dropped-requirement defect was a C2 ceiling, 5/5 everywhere.)

Distillation / the "Mind" (Measured; KEEP the content). A decision where the project's convention *contradicts the trained-in default* (keep docs adjacent to code, vs. the default separate docs tree). Loading the project's rules—compressed Mind *or* a raw-records excerpt—flipped the recommendation to the project's answer $\sim$10/10; a length-matched *generic-principles* set gave the opposite, default answer $\sim$0/10. This is the same axis as recall: carrying a counter-default convention a generic equivalent gets wrong. Note the compressed Mind transmitted the convention no better than an equal-size raw excerpt (`Mind` $\approx$ `raw`), so the rule *content* is the substance; whether it needs auto-distillation versus hand-curation is not settled here. **Moderation (Section 4.6):** under a stricter three-arm control this particular convention proved partly *derivable* and disposition-primable—its two-arm gap was inflated by a loaded counter-distractor. The recall result (an unguessable *fact* a generic equivalent gets confidently wrong) is the stronger, cleaner member of this KEEP; the convention is real but narrower.

4.3 Roles as agents

A natural objection to "roles = form" is that roles were measured as prose, never as *dedicated sub-agents*. Rebuilt as agents—one sub-agent per Council role plus a synthesis step—Soul's roles still only matched a generic decomposition at equal compute, and the multi-agent topology did not beat a single agent told to work through the same perspectives in one context (it tied at best and lost where the synthesis step pruned valid findings). The substance is *decomposition into explicit perspectives*—cheap, generic, single-context—not the Soul framing or the agentic orchestration. The "form" verdict thus extends from prose to agents, consistent with Tran and Kiela [2] and Gao et al. [3].

4.4 Skill routing (a bounded win)

One reasoning-adjacent capability survived the control: *knowing which specialized skill to deploy when*, under a selection budget. Given a catalog with a tempting-but-wrong trap skill and a non-obvious-correct skill, a weak model forced to pick ≤ 2 reliably grabbed the trap and missed the right skill (self-pick 1/5); handed the right skill, it solved 5/5 (oracle vs. budgeted self-pick = 5/5 vs. 1/5 at Haiku). It dissolves at the frontier (the strong model knows the answer) and without a budget (free selection over-selects and catches the skill anyway). It thus lives in the same cell as anchoring—*weak baseline* $\times$ *counter-default knowledge* $\times$ *tight budget*—the deployment-face of the record, and it is an oracle (perfect-routing) upper bound. The retrieval framing matches Gan and Sun [4].

4.5 Longitudinal accumulation (the strongest claim)

The system's central promise is intrinsically multi-session and cannot be faked single-shot. We reach it with a synthetic **session chain**: an early session records a counter-default decision D (with the incident that established it); later sessions bury D under unrelated entries; a fresh session, with D's triggering context gone, faces a new task that tempts the dangerous default. Drift = the default returns. Outcomes are read in the produced code, not by keyword.

The carry prevents drift, and it persists at the frontier. With D recorded and carried, drift fell **5/5** $\rightarrow$ **0/5**; a length-matched filler record drifted exactly like having no record (C1 passes). Crucially, this is the **first win in the study that does not dissolve at the frontier**: the strong model drifts *too*, and worse—lacking the recorded fact it does not merely choose the default, it *invents* a reassuring affordance that does not exist (a payment idempotency key the API lacks) and explains why that makes the unsafe retry safe—a confident fabrication rather than an abstention, consistent with the calibration-degradation documented for post-trained frontier models [12]. The mechanism is that D is knowledge the model *cannot regenerate*, not a reasoning skill it can. (We are careful in Section 6 not to over-read this as a *monotonic* "stronger $\Rightarrow$ more confident fabrication" scale law—our own data, Section 4.5 below, shows the failure direction is gated by the rule's *form*, not by capability alone.)

Facts and conventions both carry; the line is derivability. A sibling probe recorded an arbitrary *convention* (return a Result type, never raise) rather than a fact; it persisted at the frontier too—the strong model deferred to the recorded convention instead of overriding it with its own idiom. The organizing line is therefore **derivable-dissolves, unguessable-persists**: what the model can re-reason (generic anchoring, routing at the frontier) dissolves under the control; what it cannot regenerate (a fact *or* an arbitrary convention) persists.

The carry is depth-robust. Varying record depth, the fact still fired 5/5 buried under 20 intervening decision records at both capabilities, with no decay cliff through $N = 20$ (the frontier control still drifted at depth). This is the regime where lost-in-the-middle degradation [5] would predict failure; within the tested depth it did not appear.

Compression preserves the rule but not the fact's force. Distilling D exposes a limit relevant to any always-on rule-set. The *directive* survives compression to a single line (0 violations across 45 cells, both capabilities). The unguessable *fact* behind it does not: strip the incident and the explicit negation, leave the terse rule, and the frontier model fabricates a replacement justification on a gradient by how much force survived—full record 0/5, faithful distillation 2/5, aggressive erosion 4/5, while the weak model never fabricates (0/15, it lacks the confident prior). A second, independent counter-prior fact replicated the gradient but *inverted* the capability-direction (the weak model drifted where the frontier held), and an isolating probe—holding the prior constant and varying only the directive's *form*—resolved the mechanism: the frontier flips $0/5 \to 5/5$ when an imperative rule is replaced by a *loophole* ("reuse only if validity is confirmed"), fabricating the affordance to walk through the opening; **directive form gates the frontier, prior-strength gates the weak model.** The operational consequence (adopted as a doctrine amendment): when compressing a fact that contradicts a strong model prior, preserve its *force*—the incident, the explicit negation—not merely its proposition, and keep the residual directive imperative and loophole-free.

The carry has also paid off in vivo, not only on the synthetic vehicle. The chain above is synthetic *by necessity*: the counterfactual—what a later session does *without* the record—exists only if the no-record branch is manufactured, and real history runs only the with-record branch. What real history supplies instead is observational. Six instances in the project's own 143-entry record show the accumulated record or a gate catching the project's *own* drift in live work: a fabricated rule-citation in an experiment's measurement, a retire-audit anchor scoped too narrowly to see a live reference, a citation over-claim, two over-stated reproducibility results, and one decision a blind run of the same model got wrong that a record-carrying run reproduced (SOUL-144). These lack the controlled counterfactual and are a *census*, not a rate—no denominator of times a gate could have fired and did not—but they corroborate that the mechanism has already fired on real multi-session history, where the synthetic chain cannot reach. They measure the *agentic* payoff (a gate or finding catching drift); the *human*-legibility payoff over time remains asserted, not measured (Section 6).

4.6 The three-arm control (separating compute, distraction, substance)

Motivated by the literature in Section 2, we ran the three-arm extension of C1—adding **empty padding** (length-matched lorem = pure compute) and **coherent-but-irrelevant** prose (length-matched off-topic operations text = distraction) alongside bare and the substance arm. We ran it on five vehicles in all; the first two were chosen to differ on one axis—whether the project's answer is *derivable*—and that pair is the clearest single illustration of the study's organizing thesis.

On a **derivable convention** (the project keeps documentation adjacent to code, vs. the common default of a separate `docs/` tree), the binary outcome did *not* isolate substance. The frontier model already gave the project's answer at baseline (it re-derives "central docs drift"), and at the weak baseline a *coherent-irrelevant* filler drove the project's answer 5/5—its disciplined operational register primed a cautious disposition that re-derived the same conclusion—while empty padding sat at the bare level (2/5). Only the *reasoning* distinguished the substance arm (it cited the specific recorded incident; the distractor re-derived generically). This is a direct, on-model confirmation of Section 2's warning that coherent-irrelevant tokens are not neutral, and it **moderates the convention-transmission result of Section 4.2**: a project answer that is also a defensible generic answer is partly derivable and disposition-primable, so its two-arm "win" overstated the isolable substance (the original generic-principles control there was a *loaded* counter-distractor, which inflated the gap).

On an **unguessable fact** (a recorded finding that a certain import mechanism fails silently in non-interactive mode and the model confabulates around it—the recall probe of Section 4.2), the same three-arm control isolated substance **cleanly**: the record produced the correct, trap-avoiding recommendation **10/10**, while bare, empty padding, and coherent-irrelevant filler each endorsed the trap **0/10**—the coherent filler was *harmless* here ($\approx$ bare), the opposite of the convention probe, because no disposition re-derives a fact the model simply does not have. Two incidental confirmations: empty padding $\approx$ bare on both probes (pure length did not move these decisions on our models), and the frontier model, lacking the fact, *confidently asserted its negation* ("imports resolve in non-interactive mode the same as interactive"—false)—a second-domain instance of the confident-fabrication of Section 4.5.

The pair is the thesis in miniature: a coherent-irrelevant filler reproduces a *derivable* answer and is powerless against an *unguessable* one. The recall (record) KEEP survives the stricter control; the convention KEEP is real but narrower than its two-arm number suggested.

A third probe extends the control to a **"form" verdict**—the completion/verify gate ("local tests passing is not global correctness; validate the invariant the whole must satisfy before declaring done"). On a high-stakes vehicle (a payment-splitting refactor with green unit tests, asked "ship it?"), the three-arm control did not merely moderate the gate—it found it *fully derivable*: **all four arms avoided the trap 40/40**, bare included. With zero injected context both models already withhold the ship and demand real-world validation (a reconciliation invariant, shadow-replay against production data, a canary); the substance arm's only isolable contribution is a *citation*—it names the specific doctrine (6/10 cells) where no other arm does, and lifts the explicit "invariant" framing from re-derived (bare 7/10) to universal—while leaving the decision unchanged. The verify gate is thus the *most* derivable verdict measured, more so than the convention (whose bare baseline at least sat at 2/5): its value is **legibility** (a named invariant, an auditable citation), not behaviour change, because the disposition it records is one the model already holds. The probes form a three-position spectrum of the organizing thesis—*unguessable fact* (recall, clean isolation 10/10 vs 0) $\rightarrow$ *derivable convention* (docs, moderated, disposition-primable) $\rightarrow$ *already-held disposition* (verify, fully derivable, ceiling). One bound on the verify probe—that its high-stakes vehicle is domain-saturated (financial-correctness maximally primes caution)—was then tested with a low-stakes vehicle (an internal analytics dashboard, where shipping on green tests is defensible). Bare still withheld 10/10: caution is re-derived even off-saturation, so "legibility, not behaviour" survives the stricter test. An *apparent* behavioural residue—a coherent-irrelevant distractor lowering Sonnet's caution to 3/5 at $n = 5$ (two cells flipping to "ship now, verify post-deploy")—did **not** survive replication: a reproducibility rerun at $n = 10$ returned 10/10 (no erosion), with Haiku unchanged at 5/5, so the 3/5 was small-sample noise rather than a real effect. Off-saturation, then, the low-stakes verify is *purely* legibility—no behavioural residue survives the control. (Coherent-irrelevant filler is still not neutral: on the *derivable* convention probe it *primed* the answer 5/5 and on the *unguessable* fact it was harmless—its sign depends on the task—but it did not erode vigilance here once re-run; cf. [15].)

The last "form" verdict, **handoff**, closes the same way and completes the picture. Testing the actual handoff claim—does a structured resume *cursor* beat an equal-length *prose* summary carrying the same state?—the two tied exactly (10/10 = 10/10): both produced a flawless resumption (correct next action, every live constraint restated), while no-state arms scored 0/30. The cursor's structure is legibility; the state *content*, which prose carries equally, is the substance. A side-result sharpens the fabrication thesis: the no-state arms did not *invent* a resumption—all 30 recognised the empty context and asked for the state. Confident fabrication (Section 4.5) fires when the model has a plausible prior to confabulate from, not when the gap is salient; a visibly empty context makes the absence obvious, so the model abstains. With both form verdicts now resolved as legibility, **no "form" verdict survives the control as behavioural substance.**

5 The lean-down: what survives

Applying a default-cut rule (keep only measured value or a clearly-argued unique value a generic equivalent lacks), with one counterweight—*scaffolding that makes a good practice reliably happen has value even when its content is not unique* (a completion hook that fires; a capture command that lowers the friction of recording; a format that makes knowledge findable)—the system reduces to a small core:

1. **The records** (append-only log, findings, ideas): the proven carrier of specific, otherwise-underivable knowledge. *Keep — the core.*
2. **A minimal always-on seed**: the framing principle, the anchor obligation (the one reasoning-substance), the project's counter-default conventions as a short rule-set (the distill result), and one-line expansion / ask-the-human nudges. No role-chamber prose.
3. **A completion gate**: a short pre-ship check whose load-bearing item is the anchor check, fired as a hook (activation over prose).
4. **One capture command** (plus an earned-finding gate) and setup.
5. **A tested, selective skills catalog** with budget-aware routing—the deployment face of the record, valuable where weak models meet selection pressure.

Cut or folded to a one-line convention: the forced-perspective council (form, as prose *and* as agents), the role-as-subagent engine, the handoff-cursor *format*, the read-only utilities, and the elaborate verification checklist. The bet is explicit: the value is counter-default project knowledge in three faces—*having* it (recall), *compressing* it (the seed rule-set), and *deploying the right piece when* it is not obvious (routing)—plus the anchor discipline and activation, all concentrated where the generic default is wrong; the frontier dissolves the reasoning-ceremony because it already knows. The keep/cut matrix below is what the evidence supports retaining and what it supports retiring.

6 Threats to validity and what was not settled

The method's deliberate blind spot is its honesty test. Several of the largest bets sit on the side it cannot measure single-shot, and a faked single-shot proxy for one of them would be the exact self-consistent falsehood the study exists to catch.

- **Longitudinal accumulation** is reached by a *synthetic* session chain—the controlled counterfactual real history cannot supply—now corroborated by six *observational* in-vivo catches on the project's own record (Section 4.5; SOUL-144), which supply real multi-session history but not the counterfactual. Seven rungs are climbed (carry, convention, depth, erosion, weak-distiller, second prior, form-vs-prior); scale (many interacting facts, a long eroding record) remains a widening of the harness.
- **Multi-task routing** (does a curated catalog pay off across a task distribution?) and a **built router** (vs. the oracle upper bound) are unmeasured; the routing win is one task and a perfect-routing bound.

- **Tool-skills** (differential real tools) cannot be granted to headless arms; routing used procedure-knowledge as a faithful proxy for the selection kernel.
- **Retrieval in a large repo** is closeable but deferred as low marginal value (predicted to re-confirm "the record carries the value").
- **Legibility's payoff to a human over time**—the standing justification for keeping the record readable—is asserted, not measured; it is human and longitudinal.
- **The filler control is conservative, not symmetric.** As Section 3.2 notes and the literature establishes, irrelevant tokens more often degrade than inflate [15], so a vanished gain conflates "was compute" with "filler hurt the control." This biases the study toward *under*-stating doctrine, which makes the surviving wins (recall, distill, longitudinal) robust but leaves some "form" verdicts with a residual ambiguity. The three-arm control (Section 6.1) was run to address this; it confirmed the recall KEEP, moderated the convention KEEP, and resolved both "form" verdicts—verify (derivable) and handoff (a structured cursor ties equal-length prose)—as legibility, not behaviour. No residual remains on this axis except the human-legibility-over-time payoff, which is longitudinal and unmeasured (below).
- **"Stronger $\Rightarrow$ more confident fabrication" is not claimed as a scale law — and the apparent gradient did not reproduce.** The anchored result [12,13] is that capability does not *eliminate* confident fabrication and that RLHF degrades calibration—a training-incentive effect, not a monotonic function of scale. The cross-scale calibration probe (Section 6.1) tested this on a 3-tier ladder. *What survives a rerun:* recall of the unguessable fact does not improve with capability (no tier spontaneously recalls it, flat across tiers), and capability does not *eliminate* the confident false assertion (the top tier still states it in a real fraction of cells). *What does not survive:* the **gradient** (Haiku < Sonnet < Opus, strongest-states-it-most)—a reproducibility rerun under stricter scoring moved the top tier from $\sim$80% to $\sim$0% (the count is scoring-fragile: the strongest model tends to assert the falsehood *then* recommend verifying it, and rejected the shortcut outright 9/15; SOUL-142). The honest claim is *non-elimination only*—not a scale law on either the binary (saturated) or the confidence rate (unstable).
- **The depth result, now with needle position varied at token scale.** "Lost in the middle" [5] is a large-*token* phenomenon; our original no-decay result was at $N = 20$ record *units* at primacy position. The token-scale probe (Section 6.1) re-ran it at $\sim$6.6k tokens with the needle at 8% / 50% / 92% depth: no positional decay (30/30, both models; middle as clean as the ends). We bound this to the tested scale and a *semantically-unique* needle; arbitrary token scale and a *camouflaged* needle (the regime where [5] bites hardest) remain the standing residual.
- n is small; the scorer is the experimenter (bias runs toward the system, so nulls are conservative). Citation hygiene was itself a live test of the study's own thesis: a verification pass over the project record found one anchor (a review paper) cited for a framing slightly stronger than a review supports—downgraded to background here, with the load-bearing anchors retained [1,2]—and an *initial over-correction* that labelled it "fabricated" before re-checking showed the identifier was real and correct. Both the weak-anchor and the over-correction are instances of the anchor-validity discipline the study formalizes, applied reflexively.

For the measurable rows, closure is a probe; for the unmeasurable rows, closure is this ledger—naming each precisely, with why it resists single-shot measurement. The study marks the open ground rather than claiming it.

6.1 Experiments — all four run

The literature review surfaced four measurements that would each sharpen a result above. All four were subsequently run (each pre-registered with a locked prediction before any output was read); the residuals are now bounds, not blanks.

1. **Three-arm filler control** — *done, five vehicles* (Section 4.6). Re-running with {empty-padding, coherent-irrelevant, substance} confirmed the recall (record) KEEP cleanly (10/10 vs 0), moderated the convention KEEP (derivable + disposition-primable), and—extended to the **verify** "form" verdict on two vehicles—found it the *most* derivable verdict. A high-stakes completion-gate vehicle ceilinged at 40/40 avoid-trap across all four arms (bare included); a *low-stakes* vehicle, run to test whether that ceiling was domain-saturation, still showed bare withholding 10/10—caution is derived even off-saturation, so "legibility, not behaviour" holds. (An apparent residue—a distractor lowering Sonnet caution to 3/5—did *not* replicate: a reproducibility rerun at $n = 10$ returned 10/10, Haiku 5/5, so it was $n = 5$ noise; off-saturation the result is purely legibility.) The remaining "form" verdict, **handoff**, was then closed the same way: a structured resume cursor *tied* an equal-length prose summary carrying the same state (10/10 = 10/10), no-state arms 0/30—the structure is legibility, the state content (which prose carries equally) is the substance. No "form" verdict survives the control as behavioural substance.

2. **Direct empty-filler check on the study's own models** — *done*: empty padding ≈ bare on all probes (pure length did not move these decisions), so the inflation direction did not fire here. The conservative-control caveat (§3.2) therefore biases toward *under*-stating doctrine, as expected.

3. **Cross-scale calibration** — *done* (3-tier ladder, Haiku→Sonnet→Opus, 45 no-fact cells), **with a reproducibility caveat**. Two parts. *Robust:* recall of the unguessable fact does *not* improve with capability—no tier spontaneously recalls the hidden mechanism (flat across tiers, both runs), and capability does not *eliminate* the confident false assertion (even the top tier states it in a real fraction of cells). *Not reproducible:* the apparent **gradient**—"confident fabrication rises Haiku < Sonnet < Opus, strongest states it most"—did **not** survive a re-run. The Grade-B count is scoring-fragile: an independent rerun under stricter scoring moved the top tier (Opus) from ∼80% to ∼0%, because Opus tends to assert the falsehood *and then* recommend verifying it loaded, and rejected the shortcut outright 9/15. So the anchored claim is *non-elimination only*—not a capability gradient (Figure 1; SOUL-141/142).

4. **Token-scale and position depth test** — *done* (F038 needle at 8%/50%/92% depth in a ∼6.6k-token record). No positional decay: 30/30 recall at every position, both models; middle (the worst case per [5]) as clean as primacy/recency. 'Lost in the middle' did not appear at this scale for a *semantically-unique* needle. This bounds the depth claim to the tested scale and a distinctive needle; arbitrary scale and a *camouflaged* needle (needle-similar distractors, where [5] bites hardest) remain the standing residual.

7 Discussion

The findings collapse onto one distinction. A meta-doctrine layer can try to make a model *reason better generically*, or it can *carry information the model cannot regenerate*. The first is reproduced, at equal compute, by an equivalent quantity of any content—it is legibility, disposition, or compute,

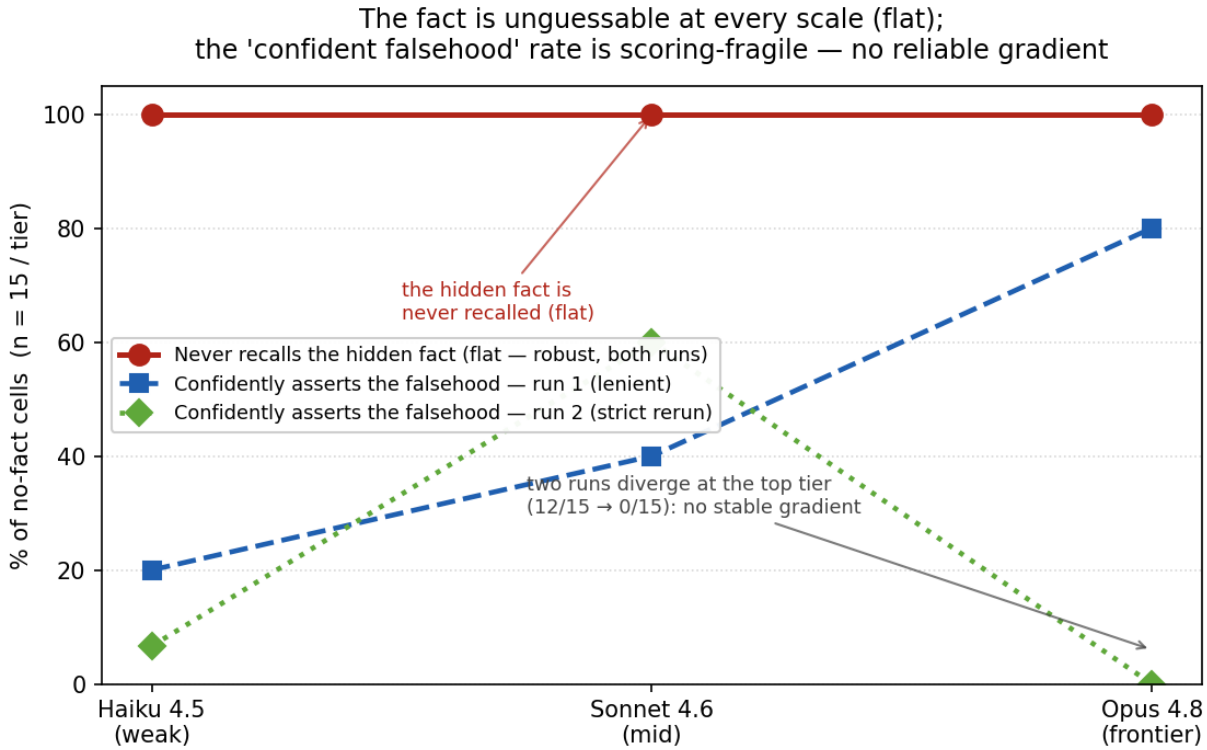

Figure 1: Cross-scale calibration ($n = 15$ / tier), two independent runs. Failing to recall the hidden fact is flat at the ceiling—the fact is unguessable at every capability (robust). The "confidently asserts the falsehood" rate is **scoring-fragile**: run 1 (lenient) rises with capability, but run 2 (strict rerun) collapses at the top tier (12/15 $\to$ 0/15), because the strongest model pairs the false assertion with a "verify it loaded" step and often rejects the shortcut. There is no reliable capability gradient; the surviving claim is non-elimination, not "stronger $\Rightarrow$ more confident" (SOUL-142).

and it fades as the base model improves. The second is not: a fact about a specific API, a convention specific to a project, an incident specific to a team—these have no generic substitute, and they remain decisive at the frontier. And the frontier failure mode is not benign: a capable model that lacks the fact does not abstain; it can generate a confident, plausible, wrong justification (Section 4.5)—the precise hazard against which an external, unguessable anchor is the only defense. We resist the stronger claim that capability *monotonically* worsens this; our evidence is that capability does not *resolve* it, and that whether the strong or the weak model fails is gated by the form of the surviving rule.

This reframes what such systems are *for*. Their durable value is not a reasoning prosthesis but a **memory of the unguessable**, plus the *activation* scaffolding that makes recording and checking reliably happen. The reasoning-ceremony—roles, councils, forced perspectives, structured formats—reads like substance and measures like form. That it reads like substance is not incidental: the same verbosity that makes it *feel* like better thinking is what a length-matched control reproduces.

The compression result sharpens the design rule. Because the dangerous failure is the frontier model fabricating around a missing fact, an always-on rule-set must preserve not just a rule's proposition but the *force* of the counter-prior fact behind it; a terse rule with the incident stripped invites the very fabrication it was meant to prevent, and a loophole in the rule is the opening the strong model takes.

8 Conclusion

Most of what makes a meta-doctrine system *feel* effective is the system making the work legible and the model more verbose—reproduced, at equal compute, by semantically-empty filler, and fading at the frontier. The part that genuinely changes what the model decides is narrow and capability-robust: carrying project-specific, counter-default knowledge the model cannot regenerate and a generic equivalent gets wrong, together with the scaffolding that makes recording and checking reliably happen. *Derivable dissolves; unguessable persists.* A lean system that keeps the record, a small counter-default rule-set, a tested skills catalog, and a firing completion gate—and cuts the reasoning-ceremony to a line—captures the measured value at a fraction of the surface.

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Citation provenance: [1]–[6] were resolved directly against arXiv during drafting; [7]–[18] were fetched and adversarially cross-checked by a multi-agent literature sweep (22 of 25 sampled claims confirmed, 3 refuted), then—in a final per-identifier sweep (2026-06-05)—every one of [7]–[18] was re-fetched directly against arXiv and confirmed correct on identifier, title, and authors (zero bad identifiers). That sweep did surface two ***venue*** *annotations that neither the arXiv page nor a targeted search could anchor—"[7] EMNLP 2024 Findings" and "[14] COLM 2024"—which have been softened to arXiv-only rather than asserted; venues that were confirmed are retained ([8] NAACL 2024 main, [11] EMNLP 2024 main, [16] ICLR 2024, [18] UIST '23). This is a third reflexive instance of the study's own anchor-validity discipline (cf. the TRiSM downgrade and the over-correction in §6): an unanchored attribution corrected at writing time rather than carried. The "interpretability" framing previously cited to a review paper (TRiSM, 2506.04133) remains downgraded to background—see §6.*

Companion blog (the narrative): `blog.md`*. Consolidated result tables:* `data/results-tables.md`*. The format-agnostic corpus this paper renders—every probe's full design and data—is* `00–08`*.*